WORKSHEET 5.6

Concavity and Points of Inflection

Unit 5: Analysing Functions

Key ideas

- The graph of f is **concave up** where $f''(x) > 0$ (equivalently, where f' is increasing) and **concave down** where $f''(x) < 0$.
- A **point of inflection** is a point of the graph where the concavity changes.
- $f''(c) = 0$ is not enough: f'' must **change sign** at c.
- A point of inflection is a point: give both coordinates when asked for the point.

Worked examples

Example 1 · A cubic

Find the point of inflection of $f(x) = x^3 + 3x^2$.

Solution. $f''(x) = 6x + 6$ changes from negative to positive at $x = -1$.

The point $(-1, 2)$**.**

Example 2 · A quartic

Find the x-coordinates of the points of inflection of $f(x) = x^4 - 12x^2$.

Solution. $f''(x) = 12x^2 - 24 = 12\left(x^2 - 2\right)$ changes sign at $x = \pm\sqrt{2}$.

$x = -\sqrt{2}$ **and** $x = \sqrt{2}$**.**

Example 3 · A bell-shaped curve

Find the x-coordinates of the points of inflection of $f(x) = e^{-x^2}$.

Solution. $f'(x) = -2xe^{-x^2}$ and $f''(x) = \left(4x^2 - 2\right)e^{-x^2}$.

$x = \pm\dfrac{1}{\sqrt{2}}$**.**

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Practice

CORE

1. Show that the graph of $f(x) = x^2 + 3x$ is concave up everywhere.

2. Find the point of inflection of $f(x) = x^3$.

3. Describe the concavity of $f(x) = \ln x$.

4. A function has $f''(x) = x - 1$. Where is its graph concave up?

EXAM LEVEL

5. Find the x-coordinates of the points of inflection of $f(x) = x^4 - 8x^3$.

6. Find the x-coordinate of the point of inflection of $f(x) = x^2 \ln x$ for $x > 0$.

7. A function has $f''(x) = (x + 2)^2(x - 3)$. Where does its graph have a point of inflection?

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8. Find the point of inflection of $f(x) = \sin x$ on $(0, 2\pi)$, and describe the concavity on each side.

Question 9 · Free-response practice

Let $f(x) = x^5 - 10x^3$.

(a) Find $f''(x)$ in factored form.

(b) Find the intervals on which the graph of f is concave up.

(c) Find the x-coordinates of all points of inflection. Justify your answer.

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Worked solutions

1. $f''(x) = 2 > 0$ for all x.
2. $f''(x) = 6x$ changes sign at $x = 0$: the point $(0, 0)$.
3. $f''(x) = -\dfrac{1}{x^2} < 0$: concave down for all $x > 0$.
4. For $x > 1$.
5. $f''(x) = 12x^2 - 48x = 12x(x - 4)$ changes sign at $x = 0$ and $x = 4$.
6. $f'(x) = 2x \ln x + x$ and $f''(x) = 2 \ln x + 3$, which changes sign at $x = e^{-3/2}$.
7. Only at $x = 3$: $(x + 2)^2$ does not change sign.
8. $f''(x) = -\sin x$ changes sign at $x = \pi$: the point $(\pi, 0)$. Concave down on $(0, \pi)$, concave up on $(\pi, 2\pi)$.
9. (a) $f'(x) = 5x^4 - 30x^2$ and $f''(x) = 20x^3 - 60x = 20x\left(x^2 - 3\right)$.
   (b) $f'' > 0$ on $\left(-\sqrt{3}, 0\right)$ and $\left(\sqrt{3}, \infty\right)$.
   (c) $x = -\sqrt{3}$, $x = 0$ and $x = \sqrt{3}$: f'' changes sign at each.

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